

Mohammed Mokhtar Soliman Salama

+20 115 036 8489 | muhammedmukhtar822@gmail.com | [linkedin.com/in/mohammedmokhtar2](https://www.linkedin.com/in/mohammedmokhtar2)

| github.com/Mukhtar82 | Portfolio

EDUCATION

Helwan National University

oct. 2022 – Jan. 2027

B.Sc. in Electrical Engineering – Intelligent Systems Major

Cairo, Egypt

- **Coursework:** Data Structures, Algorithms, Databases, Computer Systems, Machine Learning
- **Activities:** Student Union Chair – Family Committee , Head of Problem-Solving committee – Pixels Egypt

SKILLS

Languages : Python, C/C++, JavaScript (React.js, Next.js), HTML/CSS, SQL (PostgreSQL, MySQL)

AI Frameworks/Tools : TensorFlow, Keras, PyTorch, Scikit-learn, LangChain, N8N, Hugging Face

Tools and Platforms : Git, Linux, Jupyter, OpenCV, Docker, Fast API, RESTful APIs, LLMs, DL, ML, NLP, CV, RAG

EXPERIENCE

IT Engineer

July 2025 – Present

Helwan Tech Plus (Helwan National University)

Cairo, Egypt

- **Engineered and deployed a RAG-based AI assistant** using **Llama 3** and **FastAPI**, creating a centralized knowledge hub that allows students to instantly retrieve academic files and university policies.
- **Reduced administrative workload by 60%** by automating responses to routine inquiries, significantly cutting down call center volume and queuing time for university staff.
- **Optimized the student experience** by providing 24/7 self-service access to critical information, eliminating the need for physical visits or phone calls.

AI Engineer Intern

Jul. 2025 – Sept. 2025

Information Technology Institute (ITI)

Cairo, Egypt

- **Implemented Evolutionary Computing strategies**, specifically **Genetic Algorithms**, to solve complex non-linear optimization problems where traditional gradient-based methods are insufficient.
- **Leveraged Probability and Statistics** to analyze data distributions and improve model reliability, creating a strong mathematical foundation for predictive analysis.

HCIP AI Engineer Intern

Aug. 2025 – Nov. 2025

Huawei Academy (portsaid university)

Cairo, Egypt

- **Applied AI Domains:** Acquired specialized skills in **Computer Vision** (Image Preprocessing, Feature Extraction), **Speech Recognition** (GMM-HMM, DNN), and **NLP** (Text Vectorization, Language Models).
- **Full-Stack Strategy:** Trained on Huawei's "Full-stack, All-Scenario" AI portfolio, bridging the gap between theoretical algorithms and scalable industry deployment.

AI Researcher Intern

Jul. 2025 – Aug. 2025

Qayedny

Cairo, Egypt

- **Orchestrated an automated recruitment workflow** using **LangGraph**, **n8n**, and **Airtable**, integrating a custom CV parser that extracts candidate skills and auto-fills application forms to eliminate manual data entry.
- **Developed an intelligent matching system** that connects students with startups by analyzing parsed resumes against requirements, streamlining the hiring process for both parties

NLP Engineer Intern

Jul. 2025 – Aug. 2025

National Telecommunication Institute (NTI)

Cairo, Egypt

- **Optimized language models** by applying advanced preprocessing techniques (TF-IDF, N-grams) and **fine-tuning** pre-trained models to improve context understanding and similarity detection.

- **Developed a full-stack Conversational AI capstone** integrating syntax analysis, text classification, and speech processing to create responsive natural language interfaces.
- Built a back-end database service in Java and implemented a front-end UI to support future changes

Computer Vision Engineer Intern

April. 2025 – Jul. 2025

National Telecommunication Institute (NTI)

Cairo, Egypt

- **Implemented high-performance vision pipelines** for object detection and face recognition using **PyTorch**, **TensorFlow**, and **OpenCV**.
- **Enhanced model accuracy** by leveraging **Transfer Learning** on state-of-the-art architectures like **ResNet** and **EfficientNet** for complex classification and segmentation tasks.

Data Engineer Intern

April 2024 – Oct 2024

Digital Egypt Pioneers Initiative (DEPI)

Cairo, Egypt

- **Architected scalable data pipelines** using **SSIS** and **Azure**, ensuring data integrity and availability for warehousing and analytics.
- **Implemented MLOps best practices** using **MLflow**, bridging the gap between model development and production deployment.

PROJECTS

Official University Chatbot

- Engineered a production-ready RAG chatbot for the official university website using **Gemini API** and **ChromaDB**, providing students with instant, automated access to academic resources.
- Deployed the solution using **FastAPI**, **Docker**, and **Hugging Face Spaces**, ensuring scalable, 24/7 availability and low-latency responses for concurrent users.

Risk AI: Industrial IoT Safety System

- Engineered a real-time hazard detection system using **ESP32** and predictive algorithms to forecast gas leaks with 2-second latency.
- Developed a **React** dashboard visualizing live sensor streams (CO2, Temp) via high-performance Recharts.

NeuroCaption: Multimodal Image Captioning

- Designed a unified AI pipeline integrating **ResNet** (CV) and **GPT-2** (NLP) to generate semantic descriptions for input images.
- Deployed a decoupled architecture with a **FastAPI** inference engine (Dockerized) and a **Next.js** frontend on Vercel.

VitalSense: AI Health Prediction Platform

- Built a full-stack diagnostic system utilizing Machine Learning models to analyze vital signs (Heart Rate, BP, Oxygen).
- Orchestrated a **REST API** architecture connecting a Python backend to a responsive React interface for real-time predictions.

Volunteering

Head of Problem-Solving Committee

Sept. 2023 – Aug. 2025

Pixels Egypt org.

Cairo, Egypt

- Trained volunteers in algorithms and data structures; Communication and Presentation Skills
- Trained 250+ University Students all around Egypt in algorithms and data structures; organized contests and events.

Deputy Organizer

Sept. 2023 – Present

Resala Charity Organization

Cairo, Egypt

- Led medical camp logistics to serve thousands in underserved Egyptian villages.

Student Union Chair Family Committee

Sept. 2024 – Present

Helwan National University

Cairo, Egypt